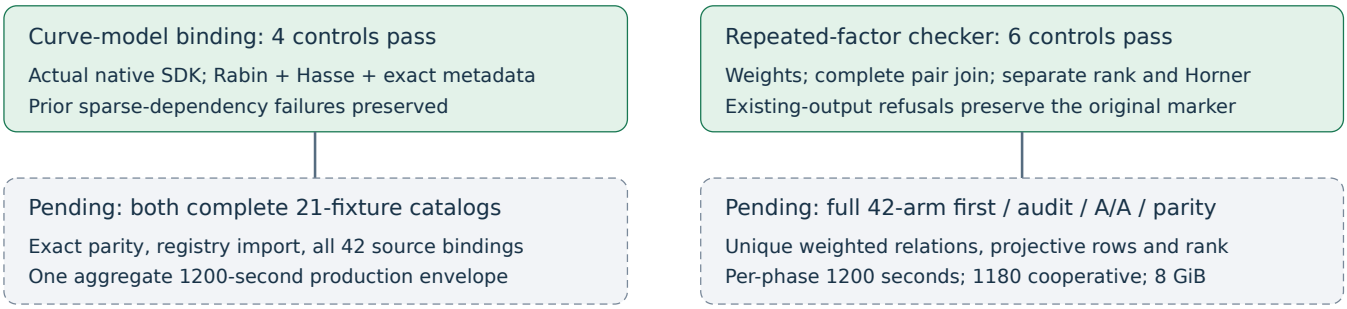
# Native controls before repeated-factor relation experiments

Observed gates; the real 42-arm and 21-fixture corpora have not run here. The completed original 189+21 and other frozen studies are retained.

| Native component | Actual outcome | Covered controls |
|---|---|---|
| Curve-model binding v3 | 4 tests pass; terminal 0 | Rabin irreducibility, unique Hasse multiple, canonical metadata, generator/field identity changes |
| Repeated-factor v1 | 6 tests pass; terminal 0 | Signed weights/cancellation, columns, rank methods, cubic multiplicities, complete pair join versus direct enumeration, q/Horner |
| Existing-output guards | Refusals 101 and 1 | Marker preserved; no new candidate or panel metadata |

## Observed native controls; full corpus outcomes remain unmeasured




Commit source/input/driver freeze before production. No favorable-case selection or automatic retries.

Controls are not a rank or speedup result. Full cold/target costs and matched references remain open.

## Preservation and exact scope

The first two SDK binding builds fail before compilation because tracked local dependencies are off disk. Existing dependency/metadata files are materialized without changing their bytes; v3 builds against the actual repository SDK. Errors and commands remain retained. One unused relation- checker argument generates a retained warning; all gates and seals pass.

All gates use one build worker and the common reservation, with 8 GiB address-space and a 1200-second aggregate envelope after queue admission. Shared-host/build clocks are diagnostic. These different suites are not comparable performance samples, so the diagram shows verification roles.

## Prospective full experiment

All 42 original phi/doubling arms remain fixed. Decode repetitions and signed cancellation, then independently enumerate every four-ordinary-slot / one-image public-point relation. Require exact weighted-set equality, the old distinct subset, retained zero/projective rows, separate fraction-free rank and full-rank public-point checks. Run first/audit/A/A/parity for all cases; failed/missing positions remain explicit. Finite-Fp-seen and non-Fp- seen sets can overlap: report their actual difference before claiming loss from pruning. Independently check every q candidate by Horner evaluation.

The identity path checks all 21 public fixtures, actual modulus/model and prime-order generator, derives the full order from its unique Hasse multiple, and calls the native SDK. Both complete processes share one 1200-second production envelope. Catalog parity, registry import, all 42 input bindings and exact EC1/UID metadata parity remain required. Preserve frozen labels.

## Evidence, identities and limits

Binding binary SHA-256: `b9b2261acde8e5cfae6ad4154c5450e3178ca98ecaadc9d688ef9e8c1ba72425`. Relation binary SHA-256: `89b64fa3943e75ef9cf88019de088c04a312c00b896129944700e28541422fab`. Original catalog SHA-256: `17ecb4c302526bc58e592466071e1c2704eaaf43a0a0ca1efe563e1bd69b3c10`.

Source protocols, native sources and exact gate/driver commands are beside this report and in ../curve_registry_binding. Raw gate metadata, binaries and refusal controls are in the versioned archive; disposable build caches are excluded. Compact receipts live under ../results/. The matched lead atlas outputs are retained; the separate KB tool is unavailable and no complete novelty audit is claimed. Commit the freeze before production.

No corpus yield, rank gain, target solve, deployed polynomial collector, complete price or breakthrough is established by these controls. Full cold/ target costs, strongest matched IC/rho/external controls and the formal field/group/parser bridges remain open. Canonical price figures, benchmark index and leaderboard are unchanged because there is no new validated performance result to promote. See BUILD_AND_CONTROLS.md for source links.